Direct Proofs

Directions: Attempt to prove as many of the following problem as you can using the direct proof method.

Problem 1: Show $\binom{2n}{2} = 2\binom{n}{2} + n^2$

Problem 2: Show $\binom{n}{1}^2 = \binom{n}{2} + \binom{n+1}{2}$

Problem 3: Show $\binom{n}{0} + \binom{n+1}{1} + \binom{n+2}{2} = \binom{n+3}{2}$ (Hint: Is there another way to write $\binom{n}{0}$ to make the calculation easier?)

Problem 4: Show $\binom{n}{k}\binom{n-k}{l} = \binom{n}{l}\binom{n-l}{k}$ with $k + l \leq n$.

Problem 5: Let n be a positive integer. Show that $\binom{2n}{n+1} + \binom{2n}{n} = \frac{1}{2}\binom{2n+2}{n+1}$

Problem 6: If n an odd integer, then n^2 is odd.

Problem 7: Let n any integer, then $n^3 - n$ is even.

Problem 8: Let a, b, and c be integers. If a divides b, and b divides c, then a divides c.

Problem 9: Let a, b, and c be integers. If $a^2 + b^2 = c^2$ is even, then at least one of the integers a and b is even.